QUANTUM COMPUTING & AI SEMINAR

QML Unleashed: Superposition & Entanglement

Exploring fundamental quantum mechanics principles and their transformations in modern Quantum Machine Learning architectures.

Academic Presentation & Framework Overview

Quantum Foundations

From Classical Bits to High-Dimensional Hilbert Spaces

Classical vs Quantum Bits

Classical Bit System

Stores binary state strictly as either 0 or 1 at any single point in time. Information storage grows linearly with bit count N, limiting simultaneous processing capabilities in high-dimensional feature spaces.

Quantum Bit (Qubit)

Exists in linear combinations of $|0\rangle$ and $|1\rangle$ via superposition. Information capacity scales exponentially as 2^N simultaneous states, offering unprecedented parallelism for machine learning tasks.

Core Quantum Principles

Superposition

Allows qubits to occupy multiple states simultaneously, giving algorithms the ability to evaluate vast solution spaces in parallel.

Entanglement

Intertwines qubit states so that one qubit directly influences another, enabling non-local quantum correlations and complex multi-variable feature modeling.

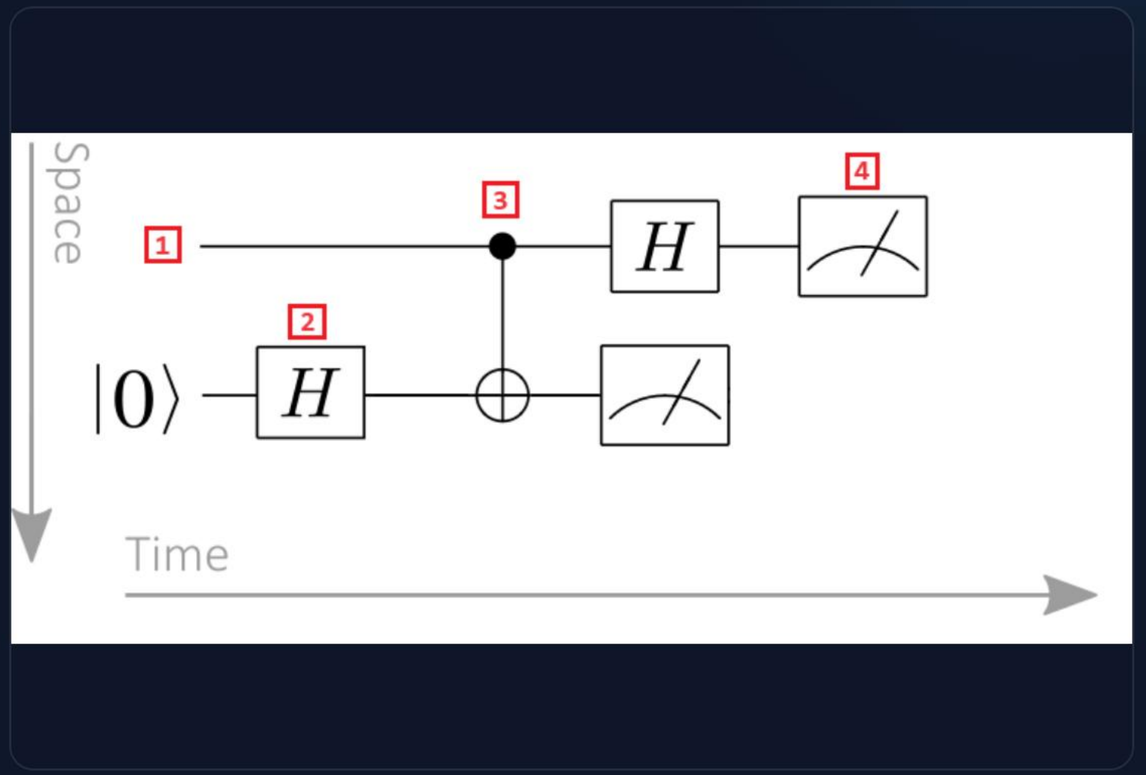
Interference

Amplifies correct computational paths while canceling out incorrect ones, driving efficient optimization in QML cost functions.

Quantum Gates & Circuits

Quantum transformations are executed using unitary quantum logic gates to manipulate state vectors:

- **Hadamard Gate (H):** Creates superposition by transforming basis states |0⟩ and |1⟩.
- **CNOT Gate:** A 2-qubit controlled gate essential for generating quantum entanglement.
- **Pauli-X Gate:** Acts as the quantum NOT gate, flipping state vectors across the Bloch sphere.



State Space Scaling

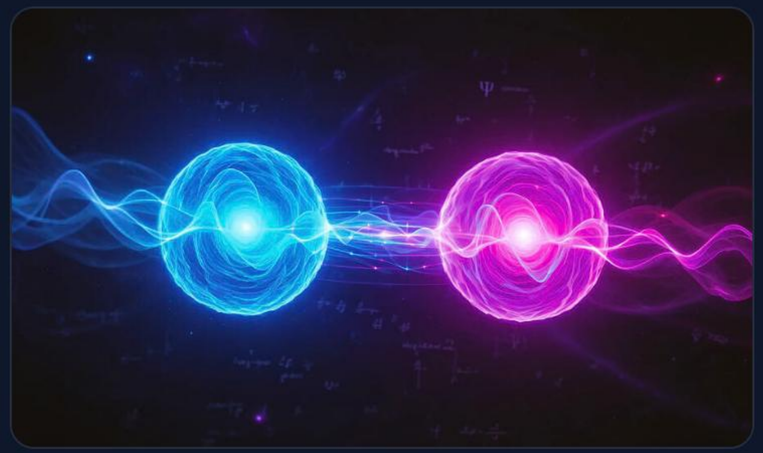
2^N

Simultaneous Computational States

Exponential Feature Expansion

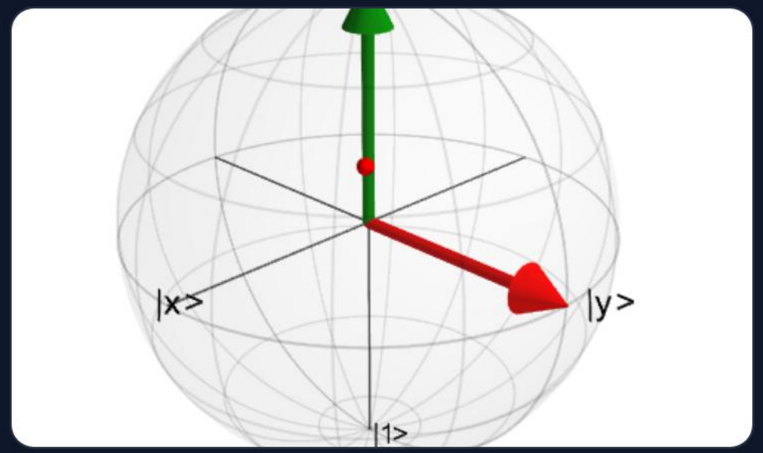
A quantum system with N qubits represents a 2^N dimensional complex vector space. With just 50 qubits, the state space exceeds the storage capacity of the world's largest classical supercomputers, providing the mathematical foundation for Quantum Machine Learning advantages.

Fundamental Quantum States



Bell States

Maximally entangled 2-qubit states essential for quantum communication protocols and QML feature maps.



Bloch Vector

Geometric representation of a single qubit state mapped onto a unit 3D sphere.



GHZ States

Multi-qubit entangled states ($N \geq 3$) demonstrating non-classical correlations in complex networks.

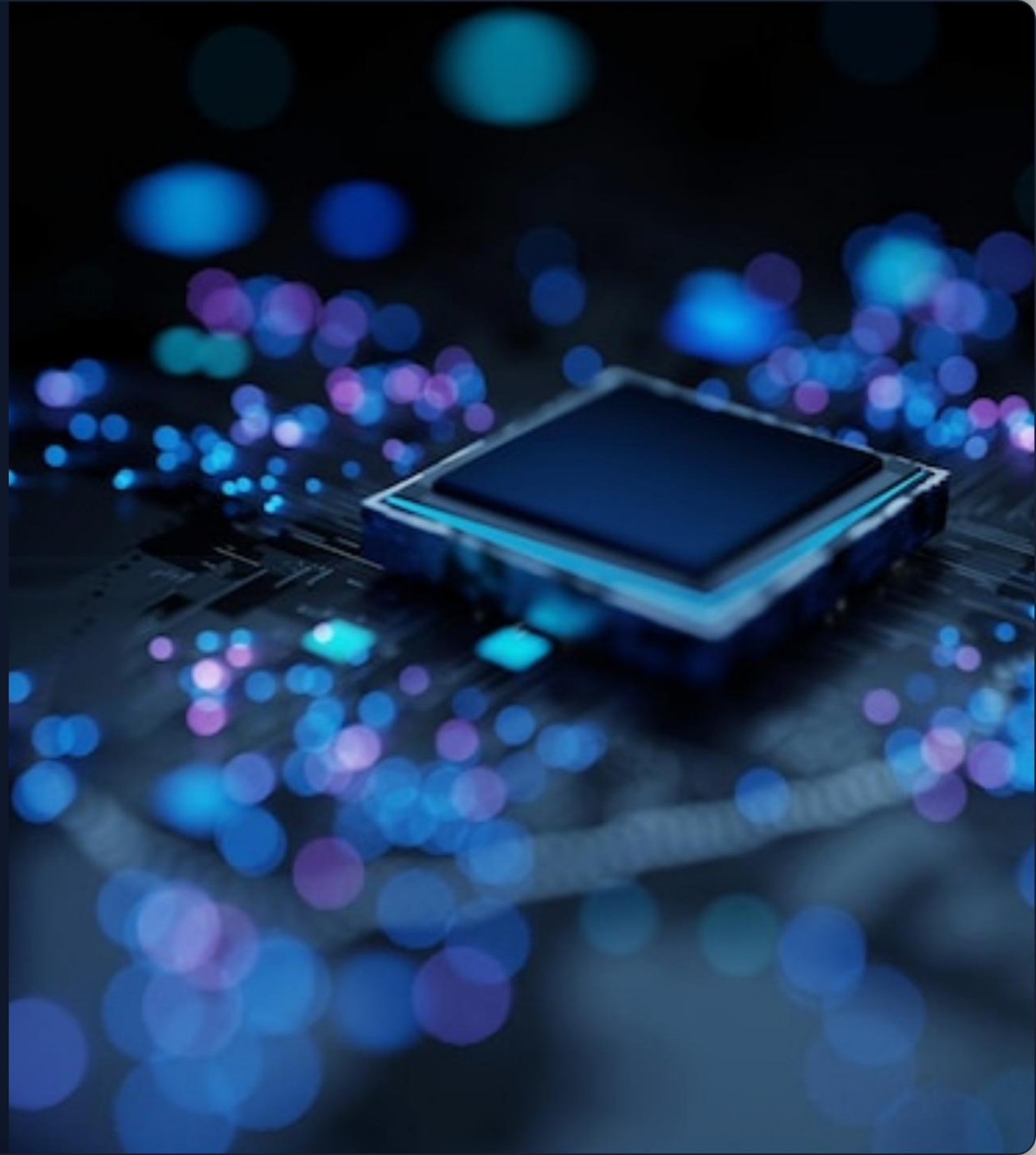
Data Encoding Strategies

- **Basis Encoding:** Maps classical binary strings directly to computational basis states $|x\rangle$. Best for discrete data lookup tables.
- **Angle Encoding:** Encodes features into qubit rotation angles (Rx, Ry, Rz gates). Highly qubit-efficient requiring N qubits for N features.
- **Amplitude Encoding:** Maps continuous vectors into quantum state amplitudes, storing 2^N data points in N qubits for exponential density.

Qiskit Implementation

Hands-On Quantum Simulation

Qiskit provides an open-source framework to build, simulate, and execute quantum circuits on real IBM Quantum processors.
By defining quantum registers, applying Hadamard and CNOT gates, and measuring state outcomes, researchers can directly observe superposition and entanglement in QML pipelines.



QML Algorithms Overview

Variational Quantum Classifiers (VQC)

Hybrid algorithms that combine parameterized quantum circuits with classical gradient descent optimizers to accurately classify complex data distributions.

Quantum Neural Networks (QNN)

Quantum equivalents of deep learning layers, utilizing entangled quantum states as non-linear weight maps for high-dimensional classification.

Feature Map Expressiveness



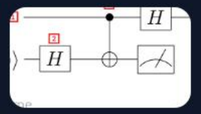
Quantum entangled feature spaces significantly enhance state expressiveness compared to classical machine learning models.

Questions & Discussion

Thank you for attending the seminar on QML Unleashed.

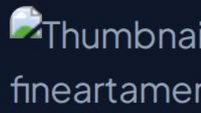
Qiskit & Quantum Machine Learning Seminar Series

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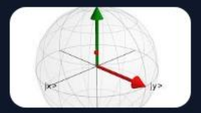
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